

Induction on the Syntax for Binary Trees

Binary trees defined by the following grammar:

$$\text{bTree} \in \textit{Tree} ::= \text{Leaf} \mid \text{Node}(\text{bTree}, \text{bTree})$$

Inductive principle for binary trees:

$$P(\text{Leaf})$$

$$\wedge$$

$$\forall \text{bt1}, \text{bt2} \in \textit{Tree}. [P(\text{bt1}) \wedge P(\text{bt2}) \Rightarrow P(\text{Node}(\text{bt1}, \text{bt2}))]$$

$$\Rightarrow$$

$$\forall \text{bt} \in \textit{Tree}. P(\text{bt})$$